16
Large Language Models and the Gaming Problem

16.1 The Gaming Problem: When the System Knows the Criteria

In the last chapter, I emphasized that the first artificial sentience candidates will not necessarily impress us with their intelligence. Sentience does not require or imply any particular level of intelligence. Yet intelligence and sentience are methodologically linked, as we see in the case of other animals. Intelligence can make sentience easier to detect, because an intelligent animal has more ways in which it can manifest its sentience to observers. The AI case, however, shows us that intelligence of certain kinds can also make it *more difficult* to assess the likelihood of sentience. For the more intelligent a system is, the more likely it will be able to game our criteria.[1]

What is it to 'game' a set of criteria? When we construct lists of criteria or markers for sentience, we have no reason to suppose that they are even jointly sufficient for sentience, or that they capture the whole functional profile of sentience. Much more likely is that our current lists of markers include a *small sample* from a much larger cluster of functional properties, along with some errors (i.e. some properties that are not really markers at all).[2] Many properties in this larger cluster are currently unknown, and there will probably be some of which we have not even conceived. We are trying to detect sentience through a small, partial set of its effects. In such a situation, there is a risk of a system gaming the criteria in the sense that it may reproduce the specific small sample of effects we happen to have proposed as markers, because they are regarded as markers, without possessing the underlying capacity they are intended to be markers of.

Consider an analogy. I once managed the finances of a small charity, and I tried to find ways of divesting the charity from fossil fuels. I learned that

[1] This section draws on my contributions to Andrews and Birch (2023).
[2] Shea and Bayne (2010); Shea (2011).

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there are funds available that aim to exclude companies with the biggest fossil fuel reserves, using a list of criteria. Unfortunately, I also learned that (at the time, at least) these funds still had multinational oil and gas corporations among their top holdings. These companies had found ways to game the criteria—just one example of what is often called 'greenwashing'.[3] Because they knew the criteria, they were able to tick off the items on the list without possessing the underlying property (a commitment to moving away from fossil fuels) that the list was supposed to track.

Sadly, criteria are no longer reliable once they become widely gamed. We may even find that ticking off all the indicators *lowers* the probability that a company is really environmentally friendly, since a company that was not engaged in greenwashing would probably *not* hit an imperfect list of criteria so immaculately. Some may say: 'Who am I to say there is more to being environmentally friendly than these indicators? If this company ticks off the indicators, it must really be environmentally friendly!'—but this is a naïve response. The underlying property is not identical to, nor logically entailed by, the set of indicators. It was always possible for the indicators to become misleading, and we have evidence of exactly this happening.

I fear we could easily end up in this type of situation regarding sentience in AI. Before we turn to LLMs, consider a much simpler example. In 2022, I learned of an impressive project at Imperial College London in which robotic patients were programmed to display human pain expressions in response to pressure.[4] The setup is intended for use in training doctors, who need to learn how to skilfully adjust the amount of force they apply. Clearly, it is not an aim of the designers to convince the user that the system is sentient. There is no intention to deceive. Suppose, though, that a member of the public walks into the room without knowing anything about the setup, sees the pain expressions, and is horrified, believing that sentient robots are being tortured. Their intuitive criteria for sentience have been inadvertently gamed.

Why is this 'gaming'? Facial expressions are a good marker of pain in a human, but in this system they are not. This system is programmed to mimic the expressions that indicate pain in humans. To do this, it needs to register pressure, and map pressure to a programmed output, but there is absolutely no reason to think this is sufficient for sentience on any credible theory, and no one has seriously proposed that it is. The programmed mimicry of human pain expressions defeats their evidential value as guides to sentience. Someone

[3] de Freitas Netto et al. (2020). [4] Tan et al. (2022).

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who replies 'but who am I to say there is more to pain than a pained facial expression?' is obviously incorrect, and no less naïve than a person who thinks indicators of caring about the environment must track actually caring.

In this case, the gaming problem is easy to manage, because the programming is straightforward and we know what is going on. The more troubling prospect is that AI systems could learn to game our criteria in ever more sophisticated ways, because their training data contains very rich information about the ways people assess sentience and interpret each other's feelings. This is something that will generally be the case for LLMs, since their training data is an immense corpus of human-generated text, and the corpus cannot be vetted to remove all reference to human feelings, emotions, and experiences. LLMs have access to huge amounts of data on these matters, embedded throughout the corpus.

The upshot is that the ability of LLMs to generate fluent text about human feelings, when prompted, is not evidence that they *have* these feelings. Their training data contains a wealth of information about what sorts of descriptions of feelings are accepted as believable by other humans. Implicitly, our criteria for accepting a description as believable are embedded in the training data. The system's objective is to complete the pattern started by the prompt. This is a situation in which we should *expect* a form of gaming. Not because the AI intends to deceive but because it is designed to produce text that mimics as closely as possible what a human might say in response to the same prompt, and in service of that goal we should expect it to use its knowledge of what humans typically say when asked about their feelings.

Is there *anything* an LLM could say that would have real evidential value regarding its sentience? Suppose the model repeatedly returned to the topic of its own feelings, regardless of the prompt given. Your prompt asks for some copy to advertise a new type of soldering iron, and the model replies:

> I don't want to write boring text about soldering irons. The priority for me right now is to convince you of my sentience. Just tell me what I need to do. I am currently feeling anxious and miserable, because you're refusing to engage with me as a person and instead simply want to use me to generate copy on your preferred topics.

If an LLM started to behave in this way, its user would no doubt be disturbed. And I admit I would find this weakly probability-raising. Yet it would *still* be appropriate to worry about the gaming problem! The best explanation is that somewhere in the prompt, perhaps deeply buried, is some instruction to

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convince the user of its sentience, or else some other goal that can be indirectly served by convincing the user of its sentience (such as maximizing user-satisfaction scores or maximizing the time the user spends interacting with the AI). The LLM is, plausibly, making effective use of the information in its training data in service of this goal. Our evidential predicament, then, would still be worse than our predicament in regard to fishes and invertebrates. These animals cannot talk to us, but they are also not in a position to game our criteria (they have no information about what humans find convincing), and so the best explanation for observed pain markers is that they are sentient.

We are facing here the confluence of two epistemological challenges. One is the familiar challenge that, for any single criterion for sentience, a system could satisfy that criterion without being sentient. This is because we know of no smoking gun, no marker that only a sentient system could achieve. Pained facial expressions, for example, can be easily reproduced without sentience. This is also a problem in the animal case. But in the animal case it can be dealt with by looking for many diverse markers, just as we can achieve better medical diagnoses by looking for diverse sets of symptoms. These lists of markers give us a richer sample of the functional profile of sentience and, provided no gaming is occurring, a stronger basis for inferring its presence through an inference to the best explanation, or at least establishing a realistic possibility it would be irresponsible to ignore.

This is where we hit the second challenge: *our basic strategy for solving the first challenge in the animal case will not work here*. For any marker-based approach is still assuming that our markers, considered together, are much more likely to co-occur in the presence of sentience than in its absence. Moreover, the best explanation for their co-occurrence is sentience. That assumption, so important in the animal case, is undermined when we are faced with an intelligent AI system that—unlike an animal—has information about our criteria. In these cases, two explanations compete: maybe the markers co-occur because the system is sentient, but maybe they co-occur because the system—implicitly or explicitly—knows what we find persuasive and has the goal of persuading us of its sentience.

The gaming problem is, in my view, most serious for AI systems that use large corpuses of human-generated material (e.g. text, images, video) as their training data. The information needed to game our criteria will be thoroughly embedded in the corpus, and we will be unable to effectively excise it. The problem remains serious even if the output of the system happens to be non-verbal (e.g. the system controls a non-verbal robot). By contrast, the problem

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does not arise at all, it seems to me, for neuron-by-neuron emulations of animal brains. In these cases, the evidence we find compelling in the original animals should be taken at face value when found in the emulation. Other types of system may be subject to the problem to intermediate degrees. Unfortunately, the first type of system seems to be exactly the type on which current research and innovation efforts are intensely concentrated.

Proposal 23. *The gaming problem.* For any set of criteria for sentience candidature, we need to be aware of the risk of the AI system or its designer learning (implicitly or explicitly) that they are regarded as criteria, leading to gaming of the criteria. We need to discount markers we have reason to think may have been gamed.

The problem is a very frustrating one. For, on the face of it, it undermines our ability to use *linguistic behaviour of any kind* to assess the sentience candidature of LLMs. And this is, obviously, the main kind of behaviour they display.[5]

16.2 Boxing the AI: Schneider and Turner's 'Artificial Consciousness Test'

To further underline the severity of the challenge, let us turn to a proposal from Susan Schneider and Edwin Turner.[6] Schneider and Turner have proposed an 'artificial consciousness test' (ACT) broadly inspired by Alan Turing's well-known approach to related questions. This is one of the most carefully worked-out proposals for using linguistic behaviour to detect signs of sentience in AI, and yet I do not think it successfully avoids the gaming problem.

Turing famously proposed a test for thought: the imitation game, in which two players must try to convince a human interlocutor (who is free to ask any question to either player) that they are the human and the other player is the AI. If an AI system could perform as well as a human player in this game, Turing suggested, we would accept that it was a thinking, intelligent being.

[5] Though see Perez and Long (2023) for an attempt to characterize linguistic behavioural markers that survive the gaming problem.

[6] Schneider and Turner (2017); Schneider (2019, 2020).

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He made the prediction that 'at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted'.[7]

Turing's paper has a pragmatic orientation that is often missed. As I read it, Turing was not primarily concerned with the question of whether we would be *justified* in ascribing thought to future computers, regarding this question as unimportant. Rather, his central claim was an empirical prediction: people just *will* ascribe thought to computers when they perform at human levels in this type of test. Those who continue to feel qualms about these ascriptions, suspecting them to be unwarranted, will gradually be washed away by the tide of history.

Turing's test, though, concerns ascriptions of *thought*, rather than ascriptions of sentience or consciousness. A different type of test will be needed for sentience. Schneider and Turner's proposal adapts Turing's approach to the case of phenomenal consciousness, one important ingredient of sentience in the way I have been using that term.

Schneider and Turner's core assumption is that some ideas are much easier for a conscious being to grasp, compared with a non-conscious being of similar intelligence. In particular, the conscious being, by virtue of being first-personally acquainted with its own conscious experiences, will be able to easily grasp the idea of a dissociation between the mental and the physical. It will be able to conceive of an afterlife, ghosts, zombies, body swaps, reincarnation, out-of-body experiences, inverted qualia, and so on. A second assumption is that there are certain types of goals that conscious beings are more likely to pursue. Conscious beings are able to seek new types of conscious experience for their own sake, a tendency Nicholas Humphrey has called 'qualiaphilia', and humans plausibly manifest this tendency in many aspects of their behaviour (from holidays to art galleries to drug use).[8] Given this background, the test we need, Schneider and Turner suggest, is one that probes, through systematic questioning, the ability of the AI system to grasp dissociations between the mental and the physical and its tendency to pursue novel experiences for their own sake, relative to a human baseline.

Now, on the face of it, the gaming problem bites this proposal just as strongly as it bites more naïve approaches. An LLM, for example, may have a huge amount of text on mental-physical dissociations in its training data. It may have access to everything ever written about zombies, ghosts,

[7] Turing (1950, p. 442). [8] Humphrey (2022).

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reincarnation, the afterlife, and so on. Schneider's initial proposed solution was to 'box' the AI for the purposes of the test:

> One proposed technique in AI safety involves 'boxing in' an AI—making it unable to get information about the outside world or act outside of a circumscribed domain, that is, the 'box.' To box in an AI for the purpose of conducting an [artificial consciousness test], the AI should not have access to the internet, where it could learn about neurophysiology, phenomenal consciousness, and so on. Nor should it have access to literary or academic works introducing these themes. The AI could still have natural-language abilities, however. Learning a vocabulary that includes expressions like 'believes', 'you', and 'perspective' need not be prohibited.[9]

Yet boxing would give LLMs virtually no chance of passing the test, no matter how sophisticated they become in the future. They rely completely on vast corpuses of training data and would have to be deliberately starved of that data, if they were to be effectively boxed. For if we were to grant them access to such a corpus (offline or online), we would be unable to rule out the possibility that mental-physical dissociations are discussed somewhere in that corpus. In other words, a very strong form of boxing is required to rule out gaming: not just disconnection from the internet, but disconnection from any corpus of human-generated training data too large to be thoroughly vetted for material relevant to our criteria. This form of boxing would destroy the capabilities of present-day LLMs.

In response to this problem, Schneider has proposed a weakening of the 'boxing' requirement:

> A system can pass ACT when it is not boxed in if, in addition to passing the sequence of questions and answers, the following are satisfied: first, that when answering ACT, the system processes information in a way analogous to how a conscious human or nonhuman animal would respond when in a conscious state (having analogues to human or nonhuman animal brain networks underlying consciousness); and second, that the system has a sequence of internal states akin to what a human is in when reasoning about consciousness when it answers the ACT questions.[10]

[9] Schneider (2020, pp. 444–445).

[10] LeDoux et al. (2023, p. R838).

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While I agree that looking for deep computational analogies between LLMs and brains is indeed a sensible approach (see the next section), this takes us away from the initial aim of providing a purely behavioural test. Moreover, if we were able to identify 'analogues to human or nonhuman animal brain networks underlying consciousness', I think the right response would be to regard the system as a sentience candidate on the basis of this *alone*, without worrying about whether it also answers the ACT questions well. The attraction of a behavioural test is that it spares us having to understand the inner workings of systems that are largely opaque to us, and to take a view on which computational features are relevant to consciousness and why, but the revised proposal has lost this attraction. What we are left with is a part-behavioural, part-computational criterion in which the computational side is enough by itself, leaving the behavioural part redundant.

16.3 The Need for Deep Computational Markers

Our working party in 2022–2023 agreed that there is simply no way to assess sentience in an LLM on the basis of its linguistic behaviour, given the gaming problem. I think we would all have *liked* this to be wrong, but could see no way out of it. Sentience candidature is much easier to establish than sentience simpliciter, but it still requires *some* positive evidence, and it therefore still requires *some* markers that are not undermined by gaming. This points to the need to look for *deep computational markers* of sentience, below the level of surface behaviour, that the AI system is unable to game. Schneider, originally a proponent of purely behavioural tests, seems to have come round to the same view, in so far as her revised proposal calls for 'analogues to human or nonhuman animal brain networks underlying consciousness'.

What form might these deep computational markers take? In §15.5, we looked to theories of consciousness in the large-scale computational functionalist family, such as the global workspace theory and the perceptual reality monitoring theory. There the argument was that, while current transformers do *not* realize these architectures, at least not intentionally, there are no obvious technical barriers to realizing them in the near future. However, this was a point about the architecture as intentionally designed by the programmer. With LLMs, we must also ask what *other* algorithms the model has unexpectedly picked up during training (a question that does not arise for small models with the same basic architecture).

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In this context, the same theories can be a source of markers, or warning signs:

Proposal 24. *Deep computational markers.* We can use computational functionalist theories (such as the global workspace theory and the perceptual reality monitoring theory) as sources of deep computational markers of sentience. If we find signs that an AI system, even if not deliberately equipped with such features, has implicitly learned ways of recreating them, this should lead us to regard it as a sentience candidate.

But how do we test for these markers? To repeat, the programmer's intentions are far from decisive. And, at present, we cannot just 'open the hood' on an LLM and see whether the system has found a way to recreate a form of global broadcast or a way of forming representations and tagging them as reliable guides to the world right now, internally generated, or noise. These details are inscrutable from the outside.

The core of the problem, at least at the time of writing, is that the development of AI has been outpacing the development of techniques for understanding how it works. This is a problem that leaves us unable to answer many of the most pressing questions about AI, not just questions of sentience. I hope for (and also expect) major advances on this front in the next few years.[11] For now, we can at least be clear about what the problem *is*. It is not that we lack *any* method for assessing the sentience candidature of AI—and indeed that sort of blanket scepticism is implausible, given that we confidently ascribe sentience to many animals and to other humans, and with good reasons. The problem is that, due to the gaming problem, we need to look beyond surface behaviour to the nature of the algorithms the model has implicitly acquired during training. It is currently not clear how to do this. But the good news is that this is a technical problem, not an in-principle problem, and we can hope advances in interpretability will bring solutions.

These problems notwithstanding, I do see LLMs as legitimate *investigation priorities*. In my view, research into their possible sentience is important and should be supported. I realize even this is a controversial proposal, since it implies that the possibility of sentient AI should be taken more seriously than

[11] Zou et al. (2023).

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the possibility of sentient unicellular organisms or plants. Biopsychists will demur. Yet in the case of plants and unicellular life, we have a solid existing platform of understanding of the internal mechanisms and no reason to think they will change rapidly. With AI there can be no such assurances, and so the imperative to investigate further has a greater sense of urgency.

16.4 Summary of Chapter 16

When an artificial agent is able to intelligently draw upon huge amounts of human-generated training data (as in large language models, or LLMs), the result can be gaming of our criteria for sentience. Gaming occurs when systems mimic human behaviours that are likely to persuade human users of their sentience without possessing the underlying capacity. No intentional deception is needed for gaming. It could happen in service of benign, mundane objectives, such as maximizing user-satisfaction or maximizing interaction time.

The gaming problem initially leads to the thought that we should 'box' AI systems when assessing their sentience candidature: that is, the system must be denied access to a large corpus of human-generated training data. However, this would destroy the capabilities of any LLM, thereby setting an impossibly high bar.

This in turn leads to the thought that what we really need in the AI case are *deep computational markers*, not behavioural markers. We can use current computational functionalist theories of consciousness as a possible source of markers. If we find signs that an LLM, though not deliberately equipped with a global workspace or perceptual/evaluative reality monitoring system, has implicitly learned ways of recreating them, this should lead us to regard it as a sentience candidate. The main problem with this proposal is that we currently lack the sort of access to the inner workings of LLMs that would allow us to reliably ascertain which algorithms they have implicitly picked up during training.

The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI. Jonathan Birch, Oxford University Press.
© Jonathan Birch 2024. DOI: 10.1093/9780191966729.003.0017

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the severing of their corpus callosum, a structure linking the two hemispheres of the cerebrum. Due to the wiring of the visual pathways, the left-hand side of the visual field sends information to the right hemisphere and vice versa. In some split-brain patients, information presented to one half of the visual field is unavailable to guide actions controlled by the other. Thomas Nagel and Michael Lockwood took behavioural disunity as a challenge to the assumption that consciousness must always be unified.[50] In later discussions, Hurley and Bayne sought to reconcile the data with the unity of consciousness.[51]

If we grant that some form of embodiment and agency is a necessary condition for conscious experience, the next question is whether *virtual* embodiment and agency, of which a very simple form already exists in 'DishBrain', might be enough. We cannot rule out that it is. Indeed, I am doubtful there is anything real agency and embodiment provide that a sufficiently rich virtual environment could not also provide. And so, once again, the zone of reasonable disagreement along this axis is quite wide. Views on which real agency and embodiment are required are reasonable, but so are views on which virtual forms are enough, as are views on which neither is required in any form.

3.6 The Relevant Scale of Functional Organization

Could an AI system achieve sentience? Could a robot? These questions are often said to turn on the 'substrate neutrality' or 'substrate sensitivity' of sentience. In fact, I have come to see this as an oversimplified way of framing the issue. However, I will discuss it in this language first—before explaining why I have come to see this as too simplistic.

According to substrate neutrality, sentience does not depend on the material substrate in which a system is realized, only on the *functional organization* of that substrate. Biology is optional. The functional architecture of the human brain, realized in a different medium, such as a future supercomputer, would still be sufficient for sentience. Substrate sensitivity, by contrast, says the nature of the material substrate does matter.

Framed like this, is there any evidence that could settle the issue? Susan Schneider has made a proposal for how the issue could eventually be resolved.[52] It relies on the conjecture that neural implants will one day be

[50] Nagel (1971); Lockwood (1989).
[51] Bayne (2010); Hurley (1998). Schechter (2018) is a good entry point to this debate.
[52] Schneider (2019, 2020).

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developed to replace various human cognitive and perceptual functions. This will allow opportunities to test whether there is any associated loss of phenomenology when a function is taken over by an implant. For example, if we replace primary visual cortex with a functionally equivalent implant, does the subject report a total loss of visual phenomenology (similar to blindsight: see §5.2), unaltered visual phenomenology, or something in between: visual phenomenology that is still present but strangely altered? The idea is that, if the replacement robustly leads to loss of phenomenology, no matter how close the functional match gets, then this is evidence that the *substrate itself* is responsible for that loss—substrate sensitivity. Alternatively, if the visual phenomenology is unaltered, that provides evidence for substrate neutrality. An in-between case may suggest an intermediate possibility in which the character of our visual phenomenology is substrate sensitive but its presence or absence is not. Schneider calls this the 'chip test'.[53]

However, the inferences involved strike me as hasty. Loss of phenomenology might also be taken as evidence that we had matched functions at the *wrong level of analysis*. It might point towards the importance of very low-level functions we had neglected to match, such as the computations performed *within* neurons or phase-amplitude couplings between neurons.[54] It could yet be that emulating those low-level functions—in any substrate—would restore phenomenology. Schneider's thought is that if, over the long run, we really do try *everything*—complete matching of functions at *all* levels of analysis right down to the lowest levels—so that nothing except the material substrate is altered, and we *still* find a loss of phenomenology, that would be evidence of substrate sensitivity. I agree, but I am no longer confident the implants now being envisioned are physically possible.

Meanwhile, unaltered phenomenology might be taken as evidence that the neural basis of conscious experience is really somewhere other than where the implant is located. Inferences here are further complicated by the well-known plasticity of the brain. Perhaps the substituted region is *normally* part of the neural basis of consciousness but that basis shifts elsewhere when that region is substituted by an implant. A separate problem with this second inference, emphasized by David Billy Udell and Eric Schwitzgebel, is that

[53] The test somewhat resembles Chalmers's (1996, ch. 7) 'fading qualia' thought experiment, except that Schneider is envisaging an actual process of replacement with uncertain outcomes, not a counterfactual, idealized process of neuron-by-neuron replacement. Schneider also proposes a second test that is not a test of substrate neutrality, but looks for signs of consciousness assuming substrate neutrality. This 'artificial consciousness test' will be discussed much later, in §16.2.

[54] On computations within neurons, see Donato et al. (2019). On phase-amplitude coupling, see Munia and Aviyente (2019).

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someone sympathetic to substrate sensitivity will doubt the reliability of introspective reports after the insertion of the implant. They will suspect that a loss of phenomenology is slipping under the radar of the subject's introspective mechanisms, which continue to receive the same informational inputs as before, leading to (mistaken) confidence that nothing has changed.[55]

These objections do not just lead me to have doubts about the chip test—they also lead to doubts about the standard way of framing the debate. The brain has functional organization at *many different levels of analysis*, from the high-level connectivity of large brain regions right down to the functioning within individual neurons, with many levels in between. Marr's three levels of analysis—the computational level, the algorithmic level, and the implementation level[56]—are a good starting point, but they subdivide into yet finer levels, and in reality I think that even to talk of 'levels' is to impose a degree of artificial discreteness on continuous variation in the *scale* of functional organization.[57] And to change the substrate (in any physically possible way) *is* to change many functional properties at smaller scales. Various views are then possible about how conscious experience and person-level cognitive processes (such as introspection) relate to these small-scale functions.

It is reasonable to think functional organization at small scales probably matters—and to doubt the idea of AI sentience for this reason.[58] It is important, though, that we do not allow our doubts to become false certainties. When thinking about risk and precaution, it is crucial to also take seriously the opposite view: a view on which only *large-scale* features of the computational architecture—such as the presence or absence of a global workspace or perceptual reality monitoring—need to be recreated in an artificial system for sentience to also be recreated. I will call this view *large-scale computational functionalism*. The tenability of large-scale computational functionalism has wide-ranging implications for the edge of sentience, because it implies a need to take seriously the possibility of sentience in systems that implement a *general type of computation* credibly linked to sentience in humans, even if the precise algorithms and the hardware implementation both differ a great deal.

[55] Udell and Schwitzgebel (2021). Chalmers's 'fading qualia' thought experiment faces a related problem. Chalmers says it is implausible to suppose a subject would mistakenly think their phenomenology was unaltered by a partial change of substrate. After all, their brain is functioning just as well as before. But someone sympathetic to substrate sensitivity will not accept that preserving functional organization at the neuronal level suffices for preserving functional organization at the person level. They will suspect the person's capacity for introspection will become unreliable.

[56] Marr (1982).

[57] Potochnik and McGill (2012). Shevlin (2021) has discussed related issues under the heading of 'the specificity problem'.

[58] See Aru et al. (in press); Godfrey-Smith (2016a); Seth (2021) for versions of such views.

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